SANKET

A Wrist-Worn ESP32-S3 TinyML System for Bangla Deaf-Aid Air-Written Gesture Recognition

Technical Disclosure & Defensive Publication

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Abstract

SANKET (Bangla: সংকেত, "signal / gesture") is a wrist-worn embedded AI system that enables a Deaf person to air-write a Bangla word with their wrist and have the device recognize it on-chip in real time, without any phone, cloud service, camera, or glove.

The hardware consists of a single ESP32-S3-DevKitC-1 (N16R8) microcontroller and a single MPU6050 6-axis inertial measurement unit (IMU) — a bill of materials under US $7. The firmware runs a quantized int8 1D convolutional neural network via esp-tflite-micro with esp-nn ESP32-S3 vector-ISA acceleration, consuming only ~17 KB of flash and ~27 KB of a 90 KB tensor arena. A novel dual-core sampling + inference architecture guarantees uninterrupted 100 Hz IMU sampling by pinning sampling to one Xtensa LX7 core and interpreter->Invoke() to the other via FreeRTOS, communicated through a mutex-protected snapshot buffer. Boot-time calibration, software IIR low-pass filtering, and a fixed 6-second one-shot recording protocol together yield robust gestures from inexpensive MPU6050 clones.

On the author's smoke-test dataset of 138 samples (49 air-written আমি + 89 idle, collected across 3 wrist positions × 3 writing styles), the trained int8 model achieves 100 % validation accuracy on 28 held-out samples. The device detects the গেস্চার in real time with sustained-detection confirmation.

To the author's knowledge, this is the first open-source Bangla air-written gesture recognition system that runs entirely on an edge microcontroller, and the first to apply the dual-core sampling-plus-inference architecture to a wrist-worn deaf-aid context.

1. Problem Statement and Motivation

Bangladesh has over three million Deaf and hard-of-hearing citizens. Existing assistive technologies for sign-language or air-writing recognition are built around:

- English/ASL-centric phone apps (do not generalize to Bangla),
- Camera-based systems (require costly setup, poor privacy, fail at night, require a free line-of-sight),
- Glove-based systems (expensive flex-sensor arrays, cumbersome, socially stigmatizing),
- Cloud ML services (require continuous internet, violate privacy, often don't support Bangla).

No prior work addresses the specific intersection of (i) Bangla language, (ii) air-written gesture recognition with a single wrist-mounted IMU, (iii) on-device inference on a microcontroller, and (iv) ultra-low cost (under $10 BOM).

SANKET fills this gap. It is designed from first principles as a deaf-aid for daily, dignified use: worn like a watch, works offline, costs almost nothing to manufacture, and preserves user privacy (no data ever leaves the device).

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2. Prior Art Review

Approach	Representative work	Limitations addressed by SANKET
ASL glove with flex sensors	Pyarali et al., Pandey et al. (various)	Expensive sensors (~$30+), social stigma, English only
Camera-based sign recognition	Google MediaPipe hands, Microsoft Kinect-based	Requires phone/PC/line-of-sight, high power, Bangla rarely supported
Smartphone IMU air-writing	Air-Writing lineage (English/digits)	Phone must be in hand; not Bangla; not wearable; needs an app
Cloud-based speech-to-sign	Various apps	Requires internet, no Deaf-to-hearing direction
TinyML wrist-worn gesture	TensorFlow Lite Micro "Magic Wand" demo	Only 4 simple non-linguistic gestures; English-centric; no on-device Bangla work

Bangla is largely absent from the TinyML literature as of this disclosure date. SANKET is believed to be the first system to combine on-microcontroller TFLite-Micro inference with Bangla air-written gestures for Deaf communication.

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3. System Architecture

3.1 Overview

```
Wrist (user wears SANKET)
   │
   ▾ air-writes a Bangla word (e.g. "আমি")
MPU6050 IMU — 6-axis (ax,ay,az,gx,gy,gz) @ 100 Hz over I²C
   │
   ▾
ESP32-S3-N16R8 (dual-core Xtensa LX7 @ 240 MHz)
   ├── Core 1 (loop)      : IMU sampling + IIR filter + ring buffer
   ├── Core 1 (loop)      : every 30 samples → mutex-guarded snapshot
   │                        → notify core 0
```

```
└── Core 0 (inferTask)  : normalize + quantize snapshot → INT8 1D-CNN
                        → esp-tflite-micro + esp-nn SIMD on S3
                        → softmax → P,<class>,<confidence> over UART
```

3.2 Hardware

Component	Specification	Role
ESP32-S3-DevKitC-1 (N16R8)	240 MHz dual-core Xtensa LX7, 512 KB SRAM, 8 MB PSRAM, 16 MB flash	Compute + inference
MPU6050	6-axis IMU, ±4 g accel, ±500 °/s gyro, I²C @ 0x68	Motion sensing
Jumper wires (4)	—	SDA / SCL / 3V3 / GND

Total BOM: ≈ US $6–7.

3.3 Firmware

Two PlatformIO Arduino projects share the same wiring:

(a) Data-logger firmware (firmware/) — Streams IMU over USB-UART (921600 baud) and performs 6-second one-shot captures on command. Protocol (host → device: T\n; device → host: H…C…L…R,START…D×600… R,STOP,600). Novel protocol detail: all recordings are fixed length (600 samples at 100 Hz = 6.000 s) with no "when to stop" trigger — the firmware counts samples. This eliminates the entire class of recording-cut bugs that plague button-triggered IMU capture during vigorous motion.

(b) Inference firmware (firmware_infer/) — Loads a quantized int8 TFLite model via esp-tflite-micro with esp-nn-accelerated kernels (CONV_2D, EXPAND_DIMS, FULLY_CONNECTED, MEAN, RESHAPE, SOFTMAX). Boot-time calibration subtracts the gyro/accel bias. IIR low-pass (α = 0.30) filters each sample. A 600-slot ring buffer rolls at 100 Hz. Every 30 new samples, the ring is snapshotted and inference runs on the other core (details in §4).

3.4 Dual-core sampling + inference (novel)

Prior single-core implementations of TFLM on ESP32 encounter a subtle pathology: interpreter->Invoke() blocks the main loop for 30–100 ms, during which no IMU samples are read. When the loop resumes, it rapid-fires catch-up samples at ≈1 ms spacing (not 10 ms), producing stutter clumps in the ring buffer. Over successive inferences, the ring buffer effectively represents 7.5 s of wall-clock time compressed into its 6-second training-distribution shape, degrading recognition of subtle gestures on all but the first inference after boot.

SANKET's firmware eliminates this by:

1. Pinning loop() to core 1 to do only sampling + ring-buffer writes.
2. Running inferTask on core 0 (FreeRTOS task, priority 1).
3. Communicating via a memcpy'd snapshot buffer protected by a xSemaphoreTake-held mutex. The critical section is ≈140 µs — a fraction of the 10 ms sample interval, so sampling is never blocked.
4. Triggering inferTask with xTaskNotifyGive from loop().

After this change, the model's performance on repeated gestures matches its performance on the first gesture after boot. On the author's dataset this restored sustained-detection reliability from "occasionally

works" to "every gesture detected".

4. Data Collection Methodology (novel)

4.1 Fixed-duration one-shot capture

Each sample is exactly 600 IMU rows (6.0 s at 100 Hz). The host GUI (collector.py, PySide6) sends a T\n trigger; the firmware auto-stops when the sample counter hits 600. This eliminates "when to release" ambiguity and defeats the mechanical-shock problem of button-held captures during table-slamming gestures.

4.2 Position × Style protocol (novel)

For each target word, samples are collected across a Cartesian product:

Factor	Values	Rationale
Position (P)	P1 natural, P2 raised, P3 low	Arm pose variability
Style (S)	S1 normal, S2 slow-careful, S3 quick-sloppy	Writing-speed variability
Repetitions	≥ 5 per (P, S) cell	Within-user variability

Target: 3 P × 3 S × 5 reps = 45 samples per word. Combined with an idle class (random/rest, 30–50 samples) the dataset forces the model to learn the gesture shape rather than any particular pose or tempo.

4.3 Air-written full-word gesture (novel)

Unlike letter-by-letter air-writing (which produces inter-letter pauses indistinguishable from random motion), SANKET's gestures are connected signature-style sweeps. Each word becomes a single fluid motion with clear start, body, and end. Empirically this yields more distinct and more repeatable IMU signatures than letter decomposition.

5. Machine-Learning Pipeline

5.1 Model architecture

```
Input        (600, 6)  float32  (per-channel normalized)
Conv1D(16, 7, s=2) + BN + ReLU  → (300, 16)
Conv1D(32, 5, s=2) + BN + ReLU  → (150, 32)
Conv1D(48, 3, s=2) + BN + ReLU  → ( 75, 48)
GlobalAveragePooling1D          → ( 48,)
Dropout(0.3)
Dense(K, softmax)               → ( K,)   K = number of classes
```

Parameters: ≈ 5 K. INT8 TFLite size: ≈ 17 KB.

5.2 Augmentation strategy

To combat small-dataset overfitting, training applies 24–30 synthetic variants per real sample:

Augmentation	Range	Simulates
Time shift	± 80 samples	Early/late gesture start inside the 6 s window
Gaussian noise	$\sigma_a \in [0.005, 0.03]$ g; $\sigma_g \in [0.5, 2.5]$ °/s	MPU clone noise
Amplitude scale (per group)	0.90 – 1.10	Larger/smaller gestures
Time warp (interpolation)	0.75 – 1.25	Fast / slow writers
Pause insertion	100 – 800 ms, p = 0.25	Thinking mid-stroke

5.3 Training

Stratified split (20 % val), Adam optimizer, sparse categorical cross-entropy, class-weighted (inverse-frequency). ReduceLROnPlateau + EarlyStopping on val_loss. Up to 80 epochs. Full INT8 quantization with a representative dataset drawn from the augmented training split.

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6. Experimental Results

6.1 Dataset

Class	Samples	Across Positions	Across Styles
ami (air-written আমা)	49	P1 / P2 / P3	S1 / S2 / S3
idle (random / rest)	89	P1 / P2 / P3	—
Total	138	—	—

6.2 Validation metrics

Stratified 20 % validation split (10 ami + 18 idle):

Metric	idle	ami	overall
Precision	1.00	1.00	—
Recall	1.00	1.00	—
F1	1.00	1.00	—
Accuracy	—	—	1.00 (28/28)

Confusion matrix (rows = true, cols = pred):

```
      idle  ami
idle  [ 18    0 ]
 ami  [  0   10 ]
```

6.3 On-device performance

Property	Value
Flash occupied by model	≈ 17 KB
Tensor arena used	≈ 27 KB of 90 KB allocated
Inference latency (ESP32-S3 @ 240 MHz, esp-nn)	≈ 30–60 ms per inference
Inference frequency	≈ 3.3 Hz (every 30 samples / 300 ms)
Sustained-detection latency	≈ 600 ms (2 consecutive high-confidence frames)
IMU sampling rate (while inferring)	100.0 Hz, uninterrupted

6.4 Live demonstration

In repeated real-time runs, air-writing আমি triggers a DETECTED,ami,0.9x message within one inference cycle after the gesture completes, and the device returns to P,idle,0.9+ state within one further 6-second window as the gesture rolls out of the sliding buffer. Multiple consecutive gestures (post-dual-core fix) all detect reliably.

7. Novelty Claims

The following elements, individually and in combination, constitute the author's original contribution and are hereby disclosed to the public:

1. A wrist-worn Bangla air-written gesture recognition system running entirely on an ESP32-S3 microcontroller with a single MPU6050 IMU, with a bill of materials under US $10.

2. A fixed-duration one-shot gesture-capture protocol (6.0 s at 100 Hz) that eliminates stop-trigger reliability failures during physically vigorous gestures.

3. A dual-core sampling + inference architecture on ESP32-S3 wherein sampling runs continuously on core 1 and TFLite-Micro inference runs in parallel on core 0, communicated through a mutex-guarded ring-buffer snapshot, specifically to prevent inference-induced sample stutter from degrading recognition of subtle motion gestures.

4. A Position × Style × Repetition data-collection protocol (3 P × 3 S × ≥ 5 reps = 45 samples/word) tailored to the within-user variability of air-written Bangla.

5. Air-written full-word connected gesture encoding (signature-style sweeps) in place of letter-by-letter decomposition, for improved distinctness and reproducibility of Bangla word gestures on a single wrist-mounted IMU.

6. A TinyML pipeline for Bangla combining per-channel normalization, 24–30× data augmentation (shift / noise / scale / time-warp / pause), a ~5K-parameter 1D-CNN, full INT8 quantization, and esp-tflite-micro/esp-nn deployment, producing a <20 KB on-chip model with sustained-detection confidence > 0.9.

7. The use of this complete pipeline as a Deaf-aid communication device for Bangla speakers, allowing a Deaf user to express words through air motion that are then output over serial for subsequent text/audio display (audio output is described as a planned extension; see §8).

8. Future Work (planned extensions)

The following are planned extensions and are also disclosed to establish their derivation from the above work:

- Audio output via DFPlayer Mini + speaker. On a `DETECTED,<word>, <conf>` event the firmware will play a pre-recorded MP3 of the word via UART-controlled DFPlayer, converting the Deaf user's silent gesture into audible Bangla speech for a hearing listener.
- Expanded vocabulary to the 10 survival words most needed by Deaf Bangladeshis: আমি, তুমি, হ্যাঁ, না, ধন্যবাদ, পানি, খাবার, সাহায্য, ডাক্তার, ভালো.
- SSD1306 OLED display for simultaneous visual output of the detected

Bangla word.

- Battery + wearable enclosure (18650 + TP4056 + buck + 3D-printed strap).
- Multi-user data collection and per-user calibration for productization.
- BLE companion app forwarding transcriptions to a phone for TTS and logging.

9. License

- Code: MIT License © 2026 Hassan Hemal
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Both licenses are irrevocable and establish this work as prior art worldwide as of the publication date (2026-10-29).

10. Suggested Citation

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End of disclosure.